Oliver Simm

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Summary

Software engineering student (**88.7% GPA**) who takes initiative and figures things out in unfamiliar territory. I deliberately joined a downtown JA team where I knew no one, choosing challenge over comfort to work alongside people with entirely different perspectives. That same instinct led me to ship production systems with no prior industry experience, digitize decades-old manufacturing processes end-to-end with minimal direction, and coordinate directly with plant staff and engineers across three continents. I am looking for a co-op at a startup or large tech company where I can take ownership of real problems, explore new domains, and grow alongside engineers who care about their craft.

Technical Skills

Languages: C, Python, Java, C#, SQL, HTML/CSS, R, M68000 Assembler
Frameworks & Platforms: .NET, Angular, FastAPI, Flask, Tailwind, Power Apps, Power Automate, Power BI, Ignition MES, Flutter, OpenCV, Sentence Transformers
Databases & Cloud: Azure, Supabase, MongoDB Atlas, Docker, DigitalOcean, Kinsta, Microsoft SQL, AI Proficiency (Claude, Gemini, OpenAI APIs)
Dev Tools: Git, GitHub/GitLab, Agile/Scrum, Pair Programming, VS Code, Trello, Clickup, WordPress

Technical Experience

Magna - Exteriors Division - (Polycon Industries) April 2026 – Present
Software Developer Co-op *Guelph, ON*

- Digitized **4 paper-based processes** up to 20 years old (Vendor Request, Stock Request, Special Shipment Request, New Material Request) as production **Power Apps/Power Automate** solutions with SQL storage, file uploads, automatic PDF form saving, approval flows, and notifications, now used by **500+ plant users per app**.
- Cut annual software costs by **$90K** by re-engineering the Vendor Request app's licensing model from Premium to Standard tier, then applied the same cost-optimization approach across all 4 apps for an estimated **$400K** in total savings.
- Deployed **4 Ignition-based MES systems** to the factory floor, including live production KPI display boards and truck-part scheduling boards used for Toyota, BMW, and Mercedes-Benz production lines.
- Owned the full project lifecycle for each application: ran **requirements-gathering sessions** and demos directly with customers, and led rollover/support from legacy paper processes to production; built **Power BI** reports for KPI tracking and process logging.
- Delivered full-stack **C#/.NET** and **Angular (Tailwind)** features supporting plant production and quality workflows.

Blue Flamingo Solutions May 2025 – April 2026
Backend Software Developer *Remote*

- Served **3 enterprise clients** by architecting a **multi-tenant RAG backend** with per-client Supabase vector tables, S3 buckets, and SentenceTransformers semantic search, achieving **3–5 second** response times.
- Eliminated re-indexing downtime via a **blue-green indexing strategy**: BeautifulSoup scraped live content into a temp table while clients queried the live table, with atomic cutover on completion.
- Increased lead conversion with a **4-phase agentic lead-capture system** tracking contact fields across turns and routing qualified leads via webhook; owned the containerized **FastAPI/Gunicorn/Nginx** backend and **CI/CD (Kinsta + GitHub Actions)**.

Junior Achievement Company Program Nov 2022 – May 2023
Vice President of Technology *Toronto, ON*

- Built a **Shopify storefront** for an 18-member student business, leading a 3-person tech team mentored by **6 pros from Scotiabank, BMO, and TD**; named **Leadership in Technology Award finalist**.

Projects

Paralegal | *Flutter, Flask, MongoDB Atlas, Gemini Flash, Sentence Transformers* | **MLH Best Use of MongoDB** May 2025

- Won **MLH Best Use of MongoDB Atlas** at GDG Guelph Hackathon on a **4-person team**, building a RAG legal-query app with city-scoped semantic retrieval indexing bylaws across **3 cities** via Atlas vector search and Gemini Flash.

Treasure Runner Game Engine | *C, Python, ctypes, Docker, Check, pytest, clang-tidy* | CIS*2750 W26

- Built a **Sokoban-style puzzle game engine** in C with graph-based room/portal navigation, exposing the full API to Python via a **2-layer ctypes architecture** (bindings + OOP wrappers, opaque pointers, C-to-Python exception hierarchy); validated with **94 C unit tests**, **85% coverage**, and **Valgrind-clean** across 2,775 allocations.

Education

University of Guelph Sept 2024 – Present
B.Computing (Honours), Software Engineering Co-op | Minor: Entrepreneurship | Area of Emphasis: AI *Guelph, ON*

- **Cumulative GPA: 88.7%** | Notable grades: Calculus I **94**, OOP **94**, Software Design III **94**, Data Structures **91**

Awards & Activities

MLH Best Use of MongoDB Atlas — GDG Guelph Hackathon 2025
Google Developer Club, Marketing Lead (Oct 2024 – Present) — Leads digital design and content teams for Guelph's largest CS community (**600+ members**)
Linda Sparling Proficiency Award — Lawrence Park Collegiate Institute, 2024 | **Founded LPCI Coding Club**